



Javier Gomez Mijangos

BSc Mathematics & Computer Science · Université Paris-Saclay

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TECHNICAL SKILLS

Programming

Python · C++ · JavaScript / TypeScript · SQL **intermediate** · Bash · HTML5 / CSS3

Formal methods

Lean — formal theorem proving (Paris-Saclay coursework)

Web & Frameworks

Next.js · React · MDX · Tailwind CSS · Web Workers · WebAssembly

ML & Data

Flow matching & flow diffusion · Generative models · Data science · Image classification (KNN)

Graphics & Math

Ray tracing · three.js / react-three-fiber · MafS · KaTeX · ManimGL · mathjs

Engineering practice

Git · GitHub Actions (CI/CD) · CSP hardening · Technical writing

EDUCATION

BSc — Double Degree, Mathematics & Computer Science

Université Paris-Saclay · France

Dual curriculum: algorithms & data structures, linear algebra, topology & differential calculus, formal proof (Lean), physics.

LANGUAGES

French — fluent

English — fluent

Spanish — fluent

Mandarin Chinese — conversational

Arabic — basic

HIGHLIGHTS

- Ships production-quality projects end to end
- GitHub: Pull Shark ×2, Pair Extraordinaire ×2, YOLO
- Open to internships & collaborations

PROFILE

Mathematics & computer-science undergraduate passionate about the theoretical foundations of maths and the practical side of software. I enjoy solving hard problems and **building things from scratch** — from a WebAssembly SQL engine to generative-AI experiments — and holding them to an uncompromising standard. Self-motivated and detail-obsessed, I take full ownership of what I build and keep raising the bar. I'm applying to **Bending Spoons First Ascent** to test myself against exceptional peers and contribute from day one.

SELECTED PROJECTS

Learning platform — “Formules Vivantes” github.com/javpato/learning-website
Next.js · TypeScript · MDX · React · WebAssembly · Python (ManimGL)

- Built an **in-browser SQL course** on a real SQLite engine (sql.js/WASM) running in a Web Worker with query timeouts and row caps, plus an automated query-validation harness.
- Implemented a from-scratch **SM-2 spaced-repetition** engine and a custom declarative animation / step-presenter framework with programmatic SVG diagrams.
- Interactive 2D/3D math widgets (MafS + react-three-fiber); ManimGL pipeline; **trilingual i18n** (fr/es/en); CSP hardening and GitHub Actions CI/CD.

Flow-diffusion generative AI

in progress

Python · Flow matching · Diffusion models

- Completed multiple courses on **flow matching and flow diffusion**, and started a project applying them to build and train generative models.

Ray tracer

personal project · C++

Rendering · Computer graphics

- From-scratch **ray tracer**: ray-object intersection, shading/lighting, and recursive reflection to render 3D scenes.

Knife-Spoon-Mon — image classification

university project

Python · Data science

- Achieved **97% accuracy** classifying kitchen utensils with a custom pipeline: a self-assembled 31-image dataset, **6 geometric features**, and a **K-Nearest-Neighbours** classifier.

EXPERIENCE

Startup shadowing — HUB71

Abu Dhabi

- Shadowed early-stage startup CEOs, observing product decisions, investor meetings, and scaling challenges first-hand.